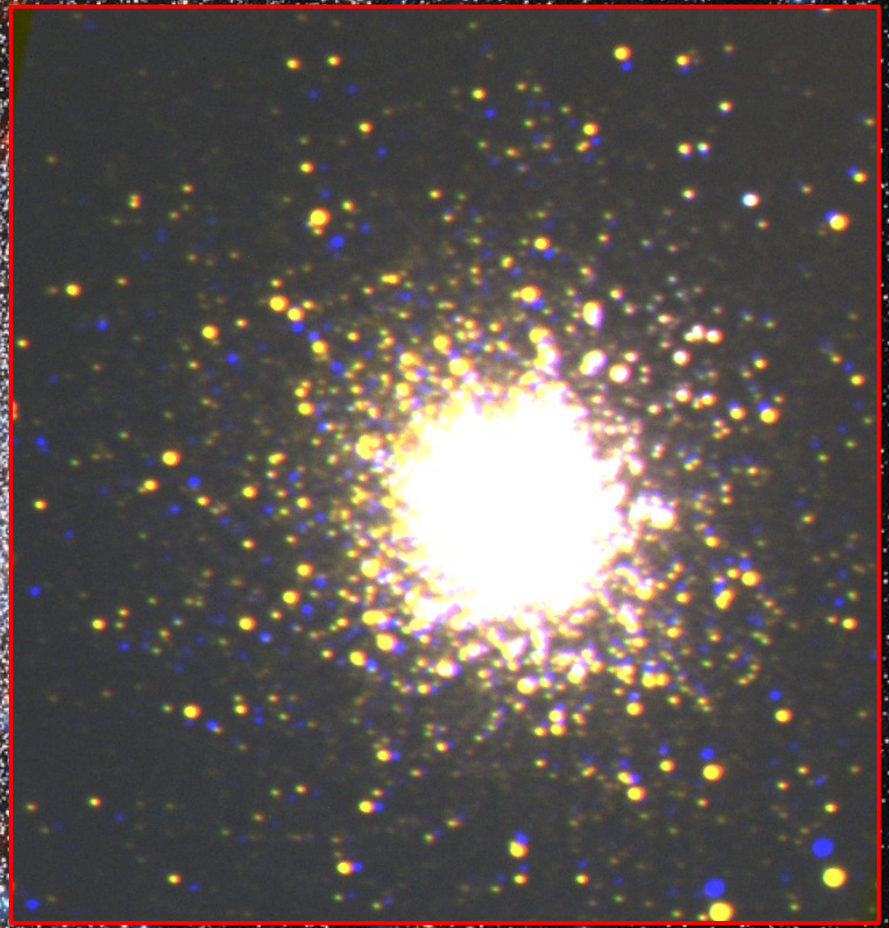
ASTR 7361
Astrophysics Laboratory
Final Project

Color Magnitude Diagram for Globular Cluster NGC 7089

by
Nageeb Zaman
Co-Investigators: Riley Dawkins, Sachin Mohandas

NGC 7089

* M2
* RA: 21h 33m 27.02s
* DEC: –00° 49′ 23.7″
* Apparent Magnitude: 6.5
* Radius: 87.3 ly
* Location: Constellation Aquarius
* Distance: 55,000 ly (17 kpc)
* Estimated age: 13 Gyr

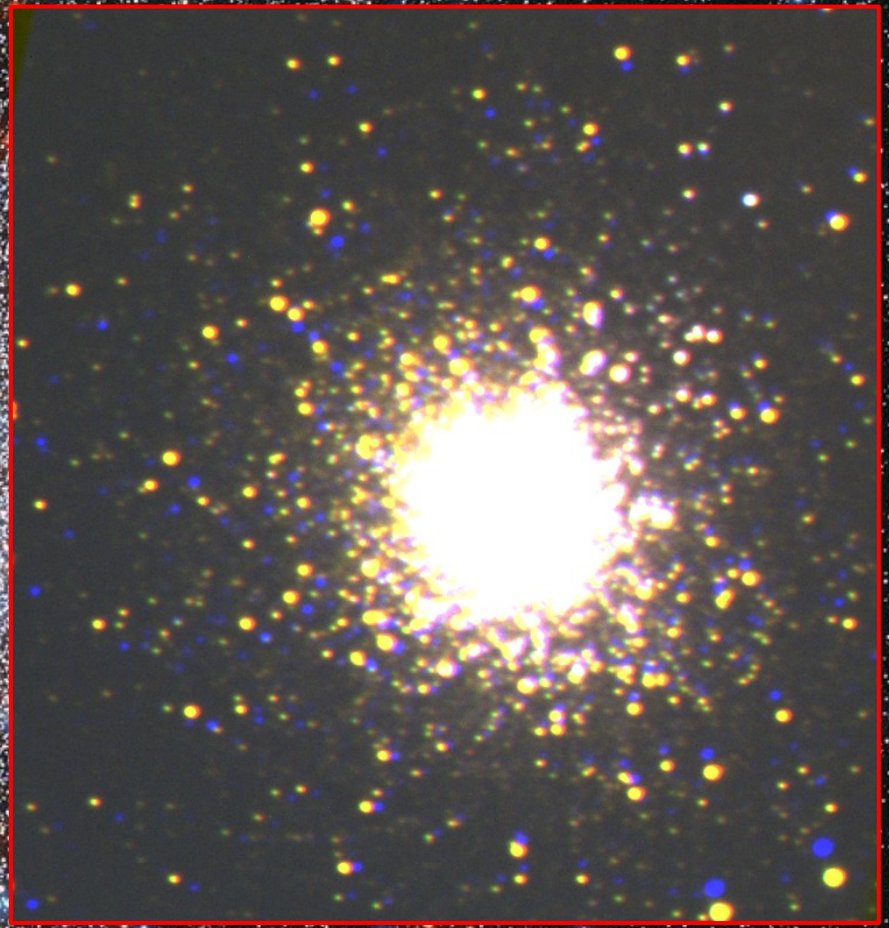


Data Collection

* 16 images (each band, 60sec)

Analysis

* Data Reduction
* Plate Solving
* Stacking

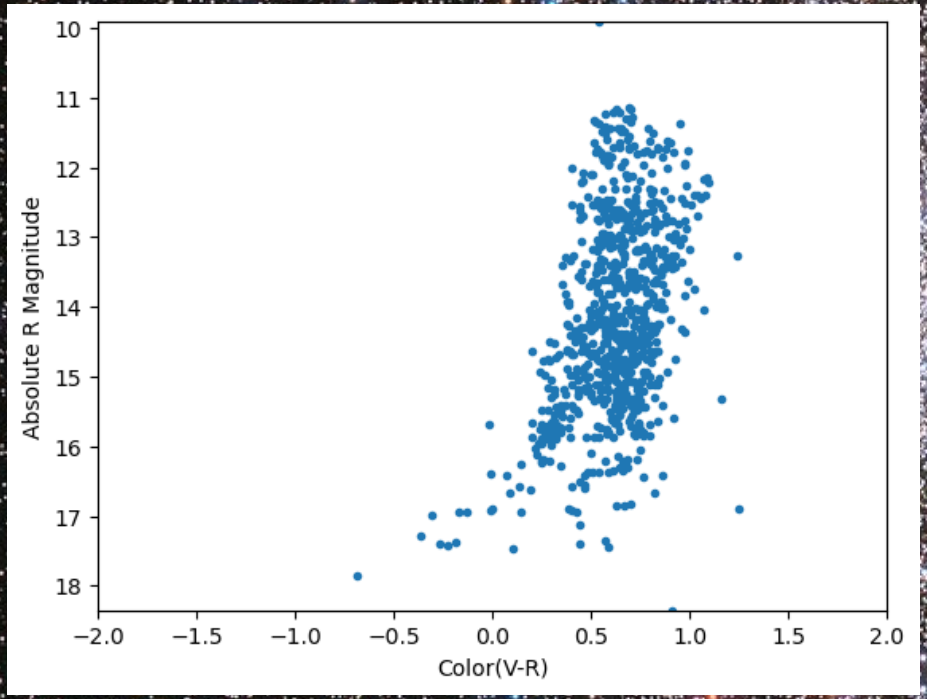


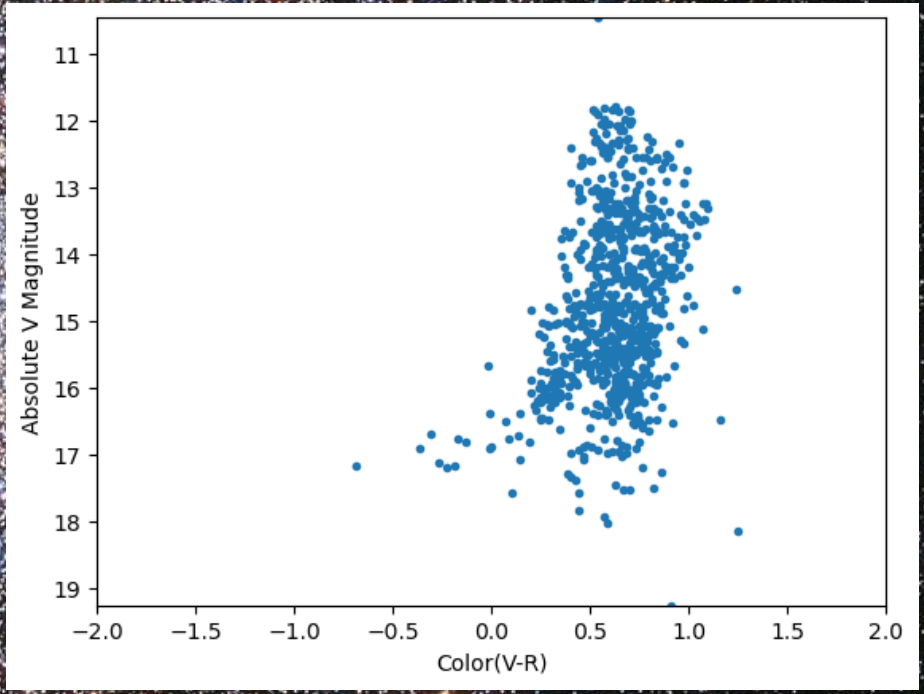
Analysis

* Photometry
  - » Point Spread Function Photometry (PSF)
    - » Python photutils.psf package
    - » Build ePSF
    - » IterativelySubtractedPSFPhotometry method

* Calibration
  - » ZMD Calibration

Color Magnitude Diagrams





Results

* **Distance:**

$$d = 10^{\mu/5+1} Parsecs$$
$$\mu = 15.5$$
$$d = 12589.25 parsecs$$

* **Age, Metallicity**

  » **Isochrone Fitting**

Conclusion

* Analysis of the collected data.
* Color magnitude diagram.
* Analysis of the CMD for various parameters of the Globular Cluster.

Thank You for your Attention